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ABSTRACT

Background: Creatine monohydrate (CrM) supplementation has been reported to increase brain creatine content by 5–15%, provide neuroprotective effects, and enhance cognitive function. However, since creatine does not cross the blood–brain barrier readily, it is recommended that individuals consume 10 g/day to promote cognitive benefit. Guanidinoacetic acid (GAA) is the precursor to creatine. Preliminary studies indicate that GAA supplementation (e.g. 2–3 g/d for 4 weeks) can increase brain creatine content to a similar degree as loading doses of CrM. However, more research is needed to assess the efficacy of long-term higher dose CrM supplementation on cognitive function and whether GAA may provide similar or additive benefits. To determine whether CrM and/or GAA supplementation affects cognitive function.

Methods: Sixty-six healthy and recreationally active adults (40.6 ± 14 years, 78.5 ± 2.1 kg, 37 females) were administered in a double-blind and randomized manner either a placebo (PLA), CrM (2 × 5 g/d), GAA (2 × 1 g/d), or CrM + GAA for 6 weeks. Prior to and following supplementation, participants completed a battery of cognitive tests (i.e. Word Recognition, Stroop Color/Word, Corsi Block, Picture Recognition, Digit Vigilance) to assess aspects of memory, reaction time, vigilance, and executive function. Baseline and week 6 data were analyzed using General Linear Model multivariate and univariate with repeated measures, mean changes from baseline with 95% confidence intervals (CI) to assess clinical significance, and percent changes from baseline using one-way ANOVA with LSD pairwise comparisons. Data are presented as mean percent changes from baseline with 95% CI (LL, UL).

Results: There was evidence that CrM + GAA improved overall reaction time (−18.1% [−34, −2], $p = 0.028$) and YES reaction time (−28.7% [−48, −10], $p = 0.006$) and tended to increase the percent of correct responses (4.4% [−0.8, 9.5], $p = 0.093$) while reducing correct reaction time (−15.3% [−33, −2.9], $p = 0.098$) compared to PLA, while YES reaction time (−29.5% [−50, −8.8], $p = 0.006$) was faster and correct NO responses tended to be higher with CrM (4.1% [−0.8, 9], $p = 0.10$) and CrM + GAA (4.8% [−0.5, 10.2], $p = 0.074$) compared to PLA in the Word Recognition test that assesses secondary memory through word familiarity and recall. In the Stroop test, which assesses cognitive function and executive control, there was evidence that CrM + GAA promoted faster overall reaction time (−18.1% [−34, −2], $p = 0.028$), correct reaction time (−11.8% [−21.5, −2.0], $p = 0.018$), incongruent reaction time (−19.9% [−34.3, −5.5], $p = 0.008$), and correct incongruent reaction time (−38.8% [−62.8, −14.9], $p = 0.002$) compared to the CrM group. However, no differences were observed compared to the PLA group in reaction time. No significant differences were observed in Corsi Block Span Score among groups ($p = 0.36$), which assesses present working and spatial memory. Similar to the Word Recognition test, there was evidence that overall reaction time on the Picture Recognition test tended to be faster with CrM + GAA compared to PLA (−8.2% [−18, 1.65], $p = 0.10$), although the percentage of correct responses was lower than PLA (−2.4% [−4.8, −.03], $p = 0.048$). There was also evidence that CrM + GAA supplementation promoted faster overall reaction time (−10.3% [−21, 0.4], $p = 0.06$), correct reaction time (−10.8% [−21.5, −0.06], $p = 0.49$), and NO reaction time (−12.9% [−28.2, 2.5], $p = 0.10$) compared to the CrM group. Finally, analysis of the Digit Vigilance test, which measures sustained attention and psychomotor

KEYWORDS
Supplement; reaction time; cognition; creatine

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speed, revealed that the percentage of correct responses was significantly higher in the CrM + GAA group compared to the GAA group (9.1% [1.6, 16.7], $p = 0.019$) with no differences observed among groups in the percentage of correct responses, correct response reaction time, or false alarm responses.
Conclusion: Results provided evidence that CrM supplementation (10 g/d) with and without GAA (2 g/d) supplementation can enhance cognitive function in healthy adults and that the more consistent positive effects were observed from supplementation of CrM and GAA. Additional research should evaluate the potential benefits of CrM + GAA supplementation in older individuals who may benefit.

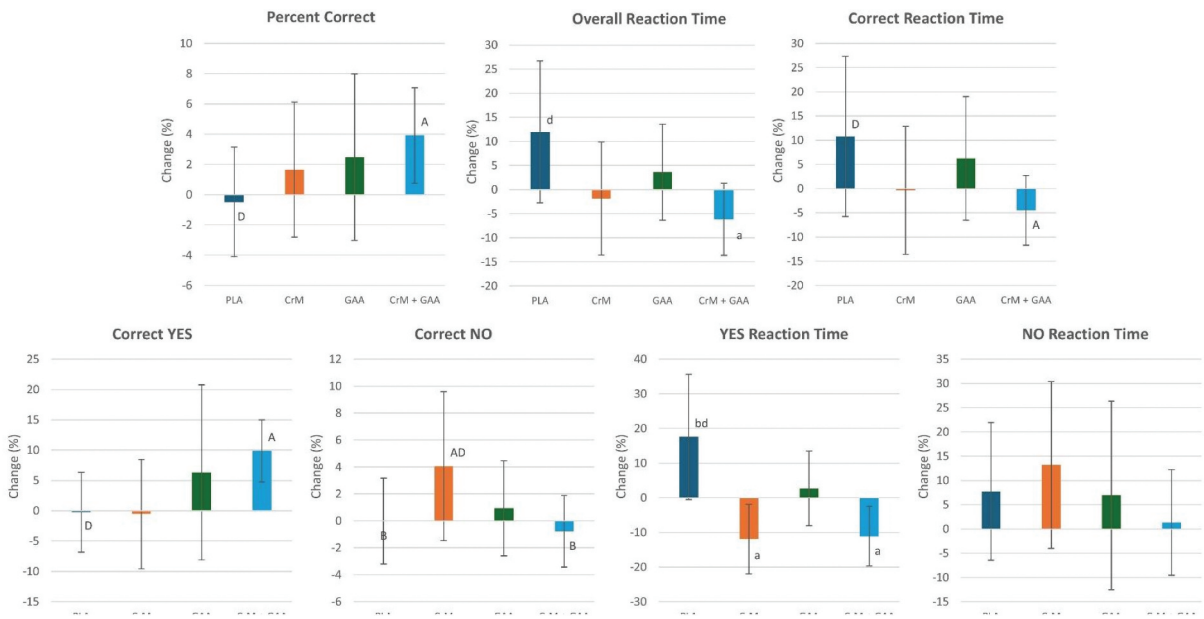

Figure 1. Word Recognition results. Data are mean changes from baseline with 95% confidence intervals for the placebo (PLA), creatine monohydrate (CrM), guanidinoacetic acid (GAA), and CrM + GAA groups. Significant differences among groups ($p < 0.05$) are shown as small letters a, b, c, and d, respectively, while capitalized letters indicate $p > 0.05$ to $p < 0.10$ differences.

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